

EE1101: Circuits and Network Analysis - Quiz 1

Name: _____

Roll No: _____

Instructions

1. The exam duration is 30 minutes, starting at 15:40. Ensure that you submit both the question paper and the cheat sheet by 16:10. Please be aware that **late submissions** may incur **penalties**.
2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.
3. Adopting unfair practices will lead to an F grade and will be reported to the disciplinary committee.

I hereby declare that I have read all the instructions carefully and will adhere to them.

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1. (5 points) Consider the DC circuit shown in Fig. 1. It is observed that the power associated with the element connected between nodes a and o is -12 W.

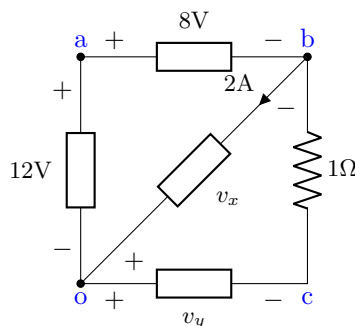


Fig. 1: Simple DC circuit (Question 1)

- (1 point) Determine the value of v_x _____.
- (1 point) Identify whether the component connected between nodes b and o is passive or active, and provide appropriate justification for your answer.
- (2 points) Comment on the type and value of the DC component connected between nodes a and b . List all potential possibilities.

- (d) (1 point) Comment on the type and value of the DC component connected between nodes o and c . List all potential possibilities.

2. (3 points) Compute the open circuit voltage, short circuit current and the equivalent resistance for the port $p - o$ in the circuit shown in Fig. 2.

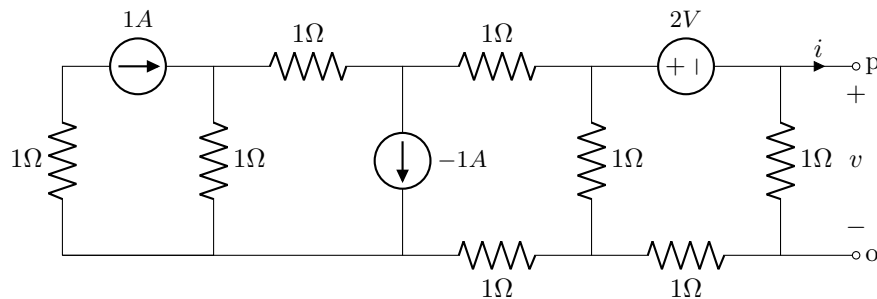


Fig. 2: Circuit for question 2

- (a) (1 point) Open circuit voltage _____
- (b) (1 point) Short circuit current _____
- (c) (1 point) Equivalent (input) resistance _____
3. (2 points) The equivalent circuit of an operational amplifier (op-amp) is shown in Fig. 3(a). Assuming the op-amp to be ideal (i.e., gain $A_v \rightarrow \infty$, input resistance $R_i \rightarrow \infty$ and output resistance $R_o \rightarrow 0$), compute the output voltage (V_l) of the circuit shown in Fig. 3(b).

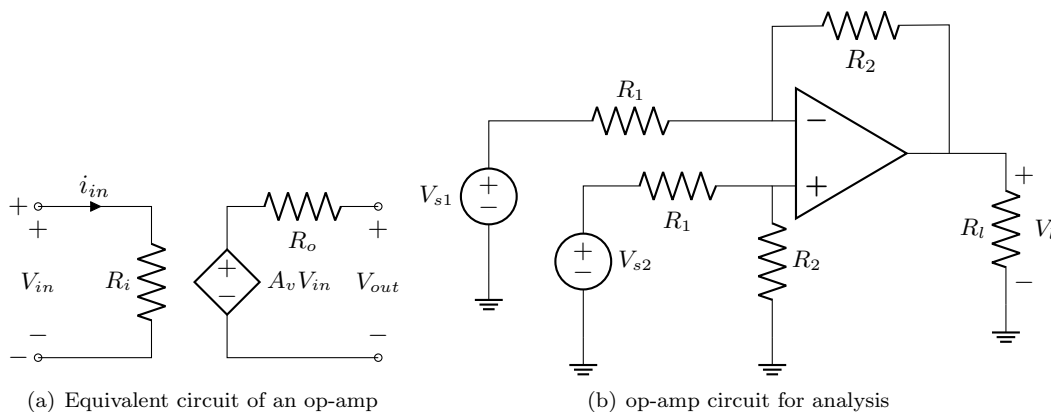


Fig. 3: Circuit with operational amplifier (Question 3)

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1. (5 points) Consider the DC circuit shown in Fig. 1. It is observed that the power associated with the element connected between nodes a and o is -36 W.

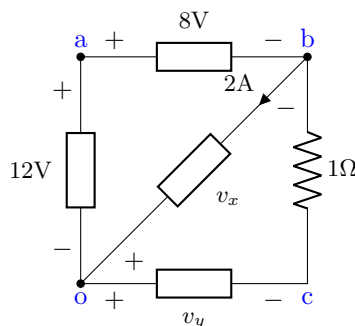


Fig. 1: Simple DC circuit (Question 1)

- (1 point) Determine the value of v_x _____.
- (1 point) Identify whether the component connected between nodes b and o is passive or active, and provide appropriate justification for your answer.
- (2 points) Comment on the type and value of the DC component connected between nodes a and b . List all potential possibilities.

- (d) (1 point) Comment on the type and value of the DC component connected between nodes o and c . List all potential possibilities.

2. (3 points) Compute the open circuit voltage, short circuit current and the equivalent resistance for the port $p - o$ in the circuit shown in Fig. 2.

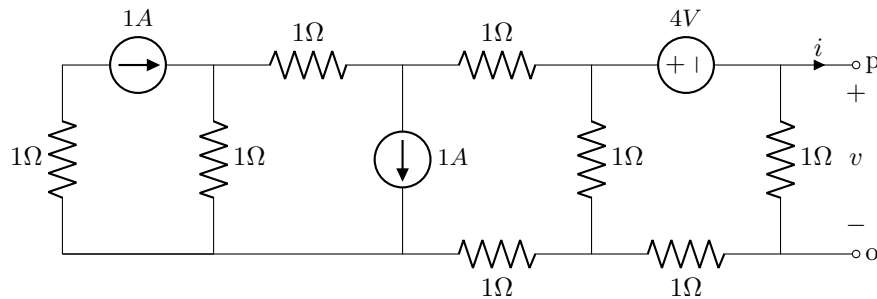
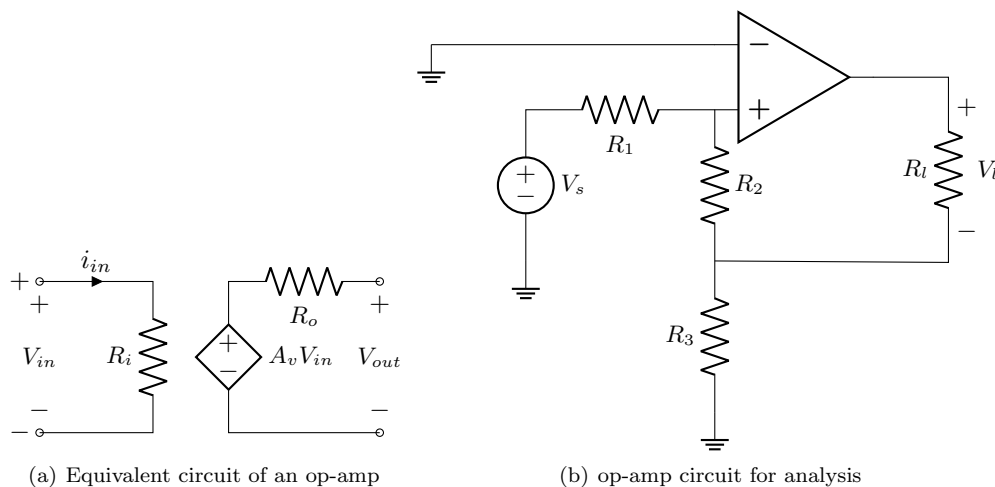


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- (a) (1 point) Open circuit voltage _____
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(a) Equivalent circuit of an op-amp

(b) op-amp circuit for analysis

Fig. 3: Circuit with operational amplifier (Question 3)

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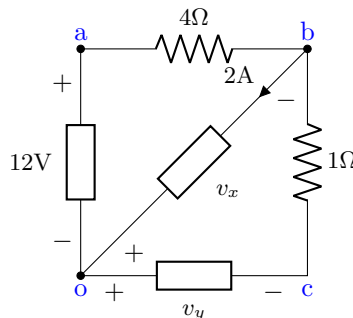


Fig. 1: Simple DC circuit (Question 1)

- (1 point) Determine the value of v_x _____.
- (1 point) Identify whether the component connected between nodes b and o is passive or active, and provide appropriate justification for your answer.
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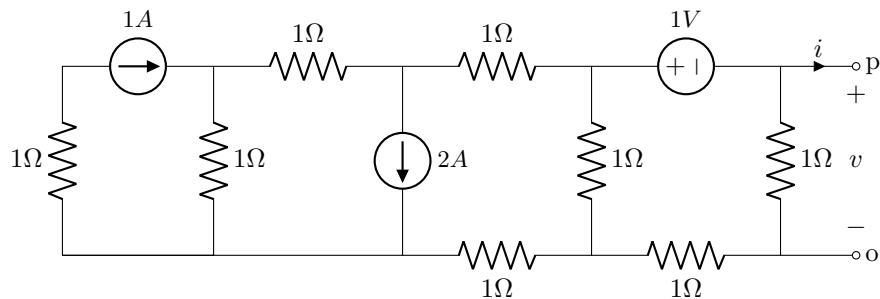


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- (a) (1 point) Open circuit voltage _____
- (b) (1 point) Short circuit current _____
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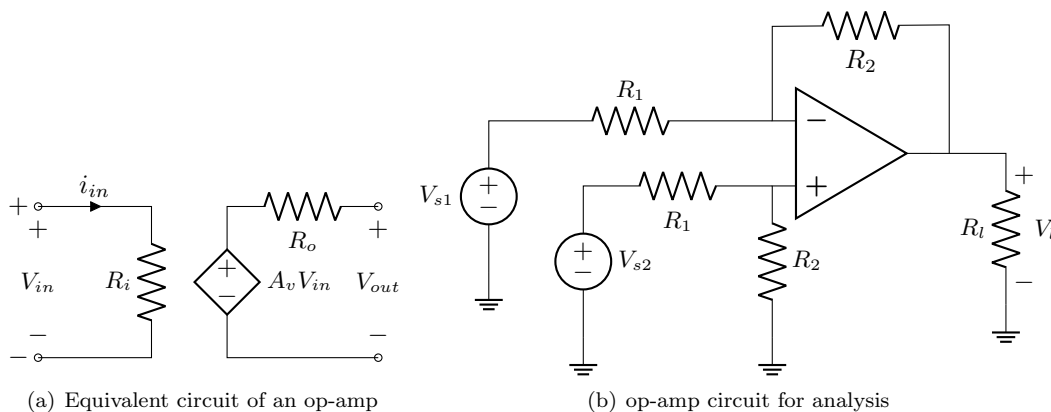


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1. (5 points) Consider the DC circuit shown in Fig. 1. It is observed that the power associated with the element connected between nodes a and o is -48 W.

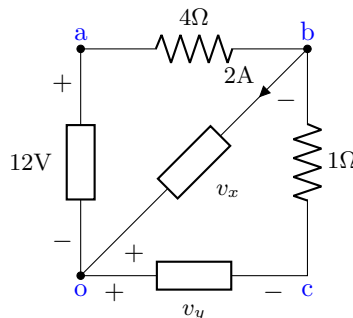


Fig. 1: Simple DC circuit (Question 1)

- (1 point) Determine the value of v_x _____.
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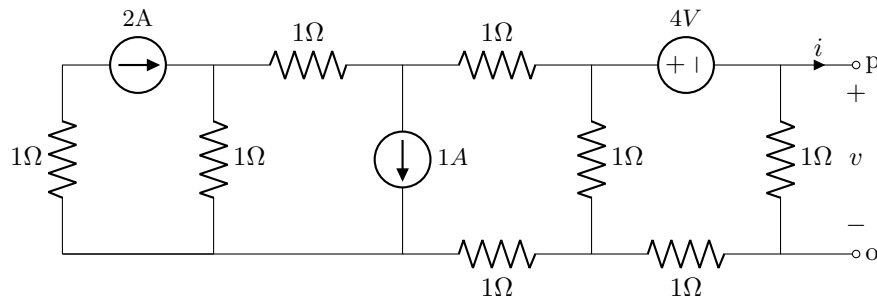


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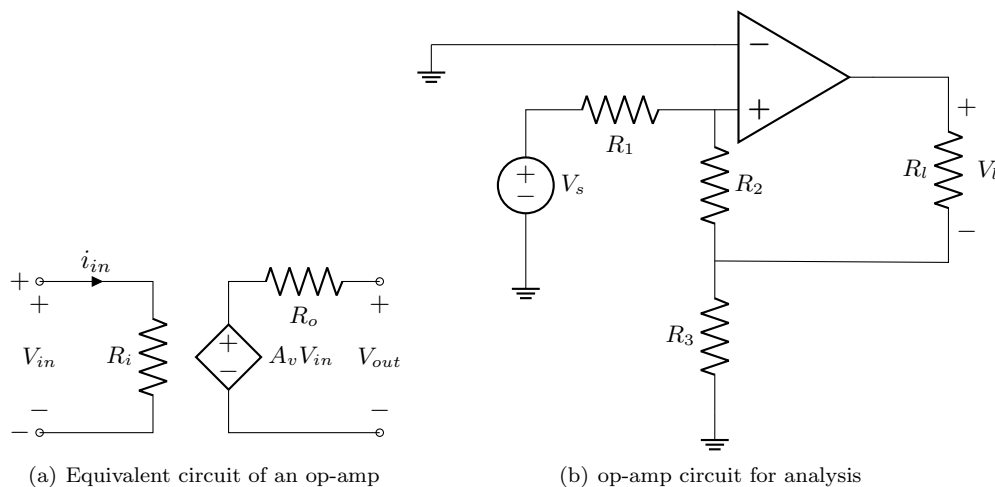


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